LaB :03

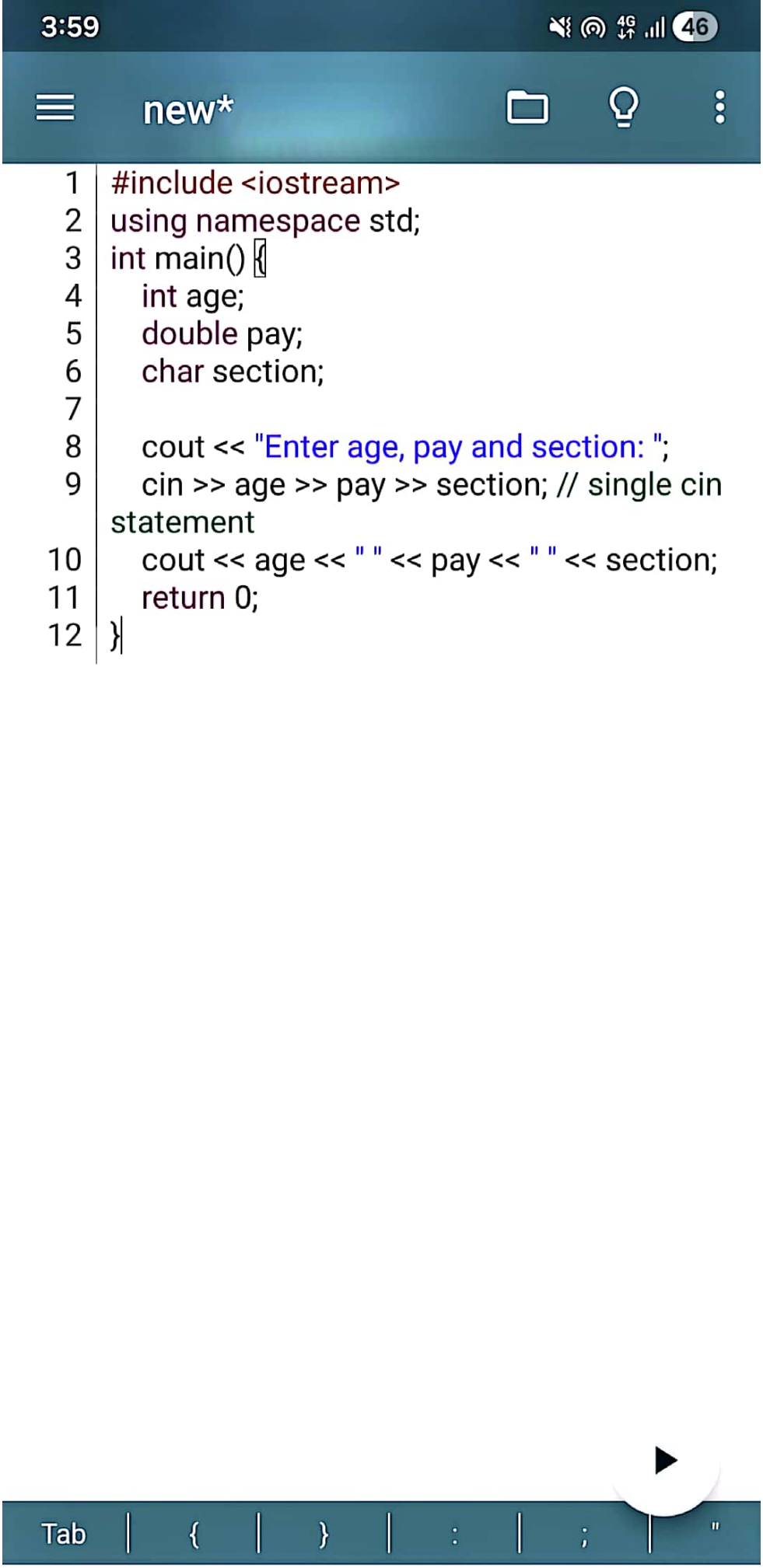
Task 1: single cin Statement

```
#include<iostream>
using namespace std;
int main( ){
    int age;
    double pay;
    char section;
    cout<<"Enter age, Pay and section :";
    cin>>age>>pay>>section;
    cout<<age<<" "<<pay<<" "<<section;
    return 0;
}
```

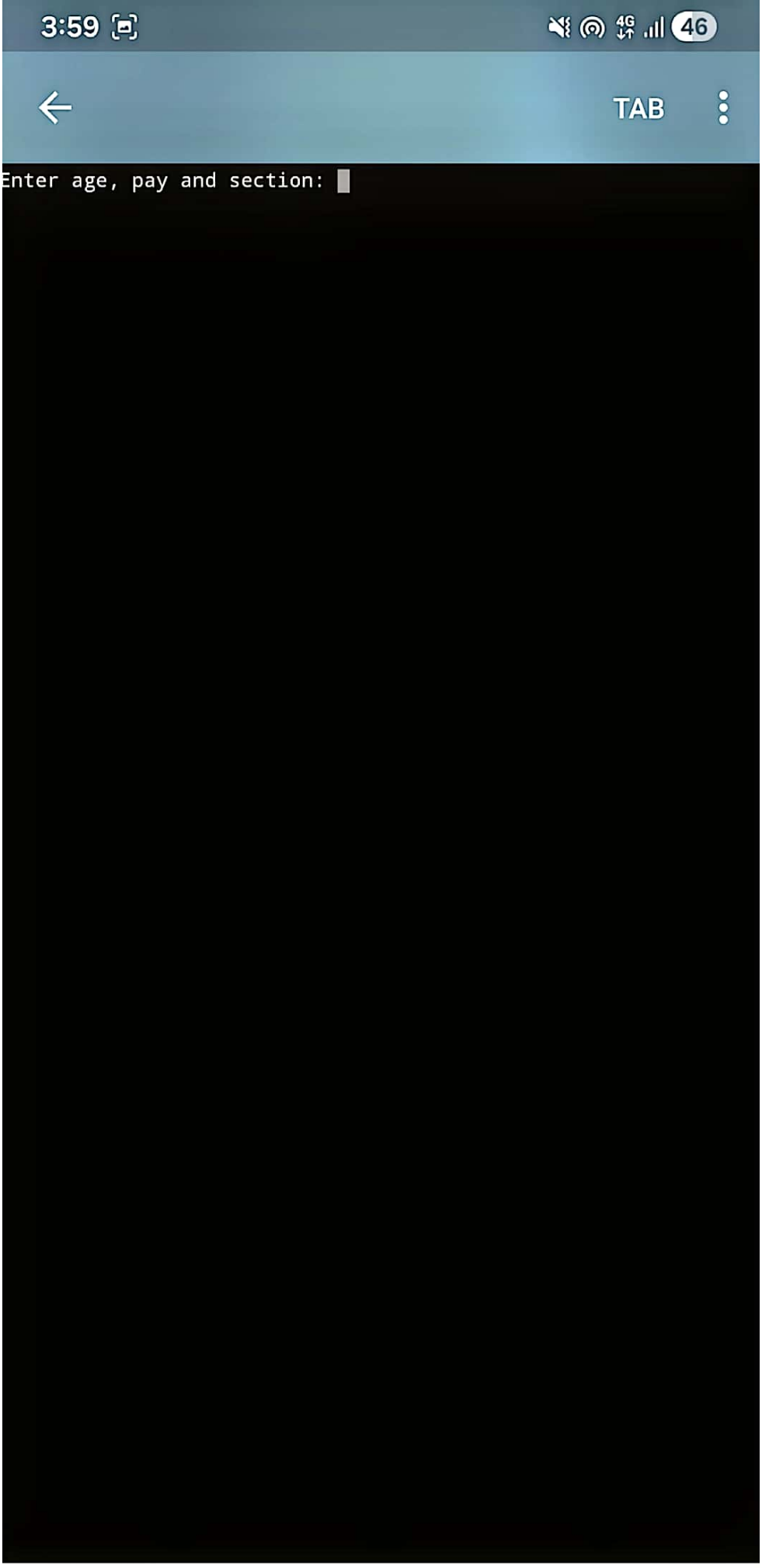
output

Enter age, Pay and section

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Task 2: Values

Expression	value
28/4-2	5
6+12*2-8	22
4+8*2	20
6+17%3-2	6
2+22*(9-7)	46
(8+7)*2	30
(6+7)%2-1	0
12/(10-6)	3
(19-3)*(2+2)/4	16

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Task:3 option c is correct

units is promoted to a float, mass
remains a float, and the result of
mass* units is promoted to a
double to be stored in weig



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Task:4

Given:

```
int a, b = 2;
float c = 4-2;
a = b* c;
```

Answer: (B. 8)

Explanation:

1. b — c means 2 — 4-2 = 8-4. The result is 8-4 of type float.
2. This float value 8-4 is assigned to a, which is an int.
3. when a float is assigned to an int, the decimal part is truncated (removed), not rounded.
4. so 8-4 becomes 8.

Therefore, the value stored in a will be 8.

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Task 5:

Given: unitsEach = qty / salesReps;
where qty and salesReps are both int.

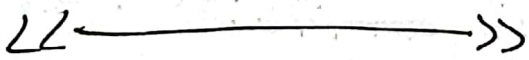
Problem: This will perform integer division, so decimal part will be lost.

Solution using Type Cast:

To avoid integer division, we need to cast one operand to float or double.

Correct Answer:

unitsEach = static_cast<double>(qty)/salesReps;



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Task 6: Math Tutor Program (rand)

```
#include<iostream>
#include<cstdlib>
#include<ctime>
using namespace std;
int main( ){
    srand(time(0));
    int num1=rand( )%900+100;
    int num 2=rand( )%900+100;
    cout<<"  "<<num1<<endl;
    cout<<"+ "<<num 2<<endl;
    cout<<"------"<<endl;
    cout<<"\n Press Enter tos ee answer...";
    cin.get( );
    cin.get( );
    cout<<"\n "<< num1<<endl;
    cout<<"+  "<<num2<<endl;
    cout<<"--------"<< endl;
    cout "  "<<num1+num2 <<endl;
    return 0;
}
```

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output of Task 6

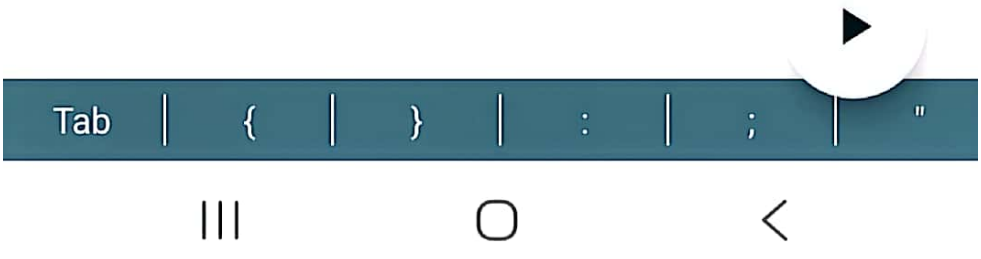
```
  846
 +204
-------
```

Press Enter to see answer...

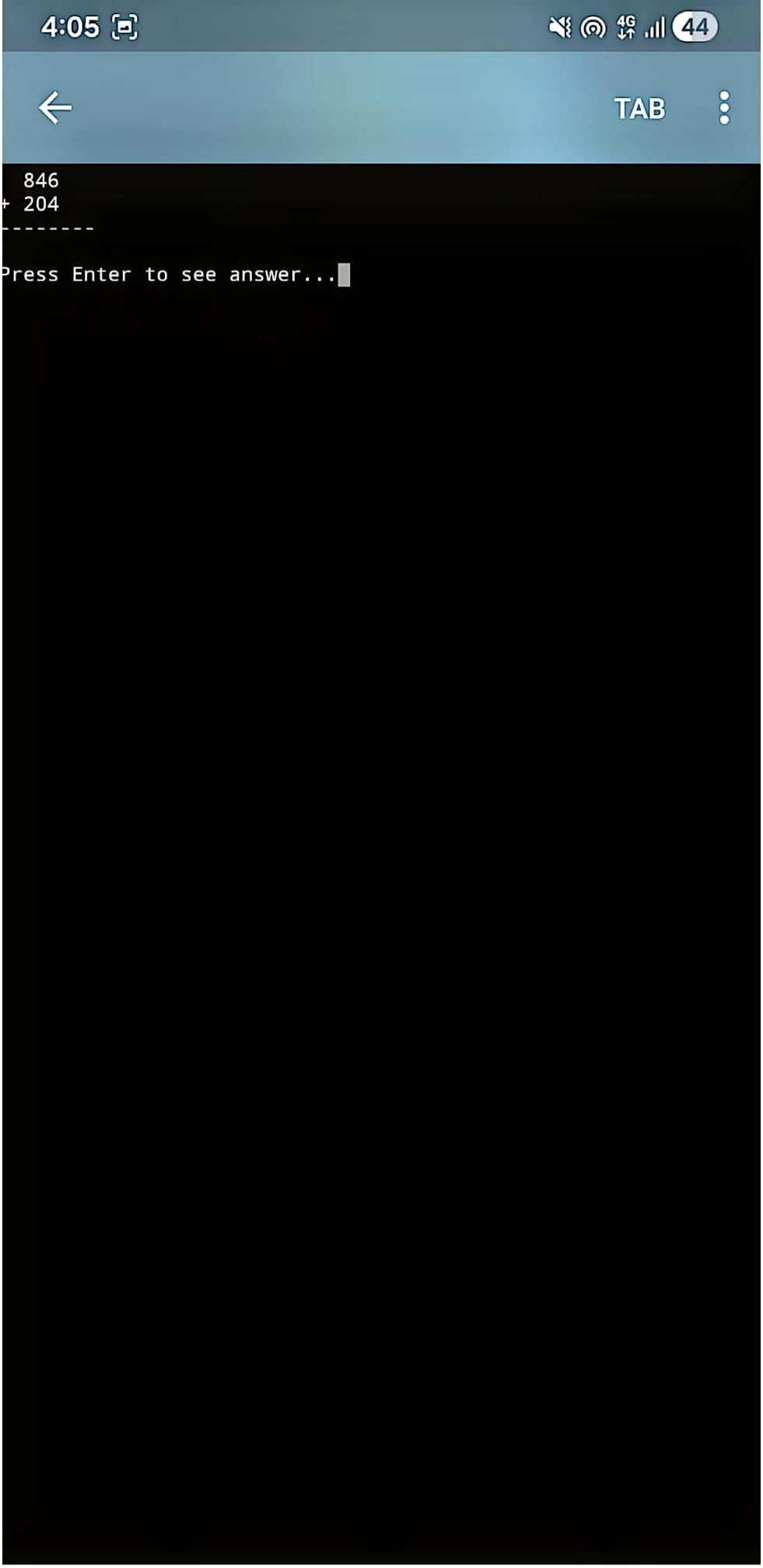
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```
1  #include <iostream>
2  #include <cstdlib> // for rand()
3  #include <ctime>   // for time()
4  using namespace std;
5
6  int main() {
7      srand(time(0));
8      int num1 = rand() % 900 + 100;
9      int num2 = rand() % 900 + 100;
10
11     cout << "  " << num1 << endl;
12     cout << "+ " << num2 << endl;
13     cout << "-------" << endl;
14
15     cout << "\nPress Enter to see answer...";
16     cin.get();
17     cin.get();
18
19     cout << "\n  " << num1 << endl;
20     cout << "+ " << num2 << endl;
21     cout << "-------" << endl;
22     cout << "  " << num1 + num2 << endl;
23
24     return 0;
25 }
```



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Lab No: 4

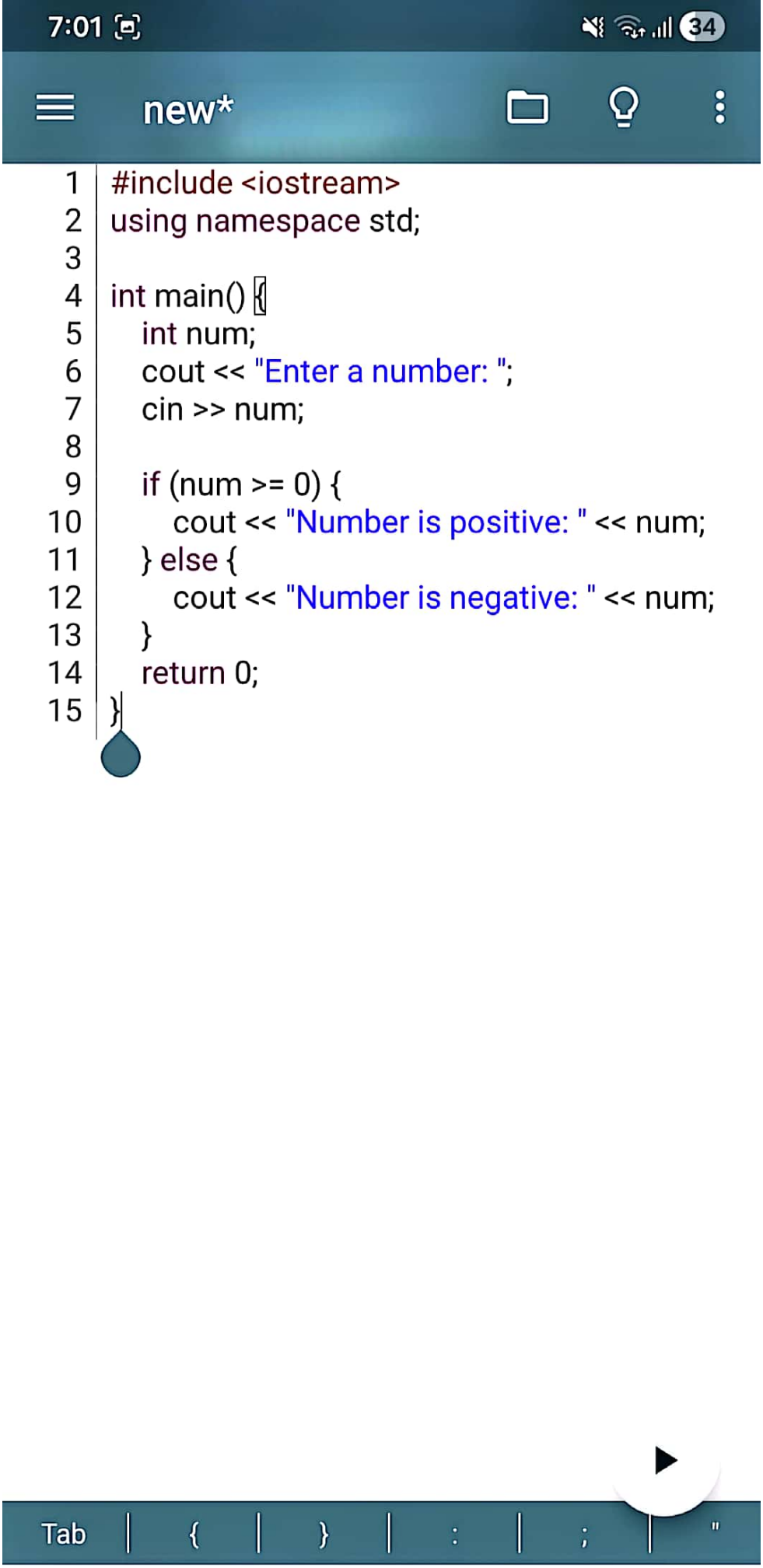
Task 1:- Posive/Negative

```
#include<iostream>
using namespace std;
int main( ){
   int num;
   cout<<"Enter a number: ";
   cin>>num;
   if(nums=0){
        cout<<"Number is Positive:"<<num;
   } else {
        cout<<"Number is negative:"<<num;
   }
   return 0;
}
```

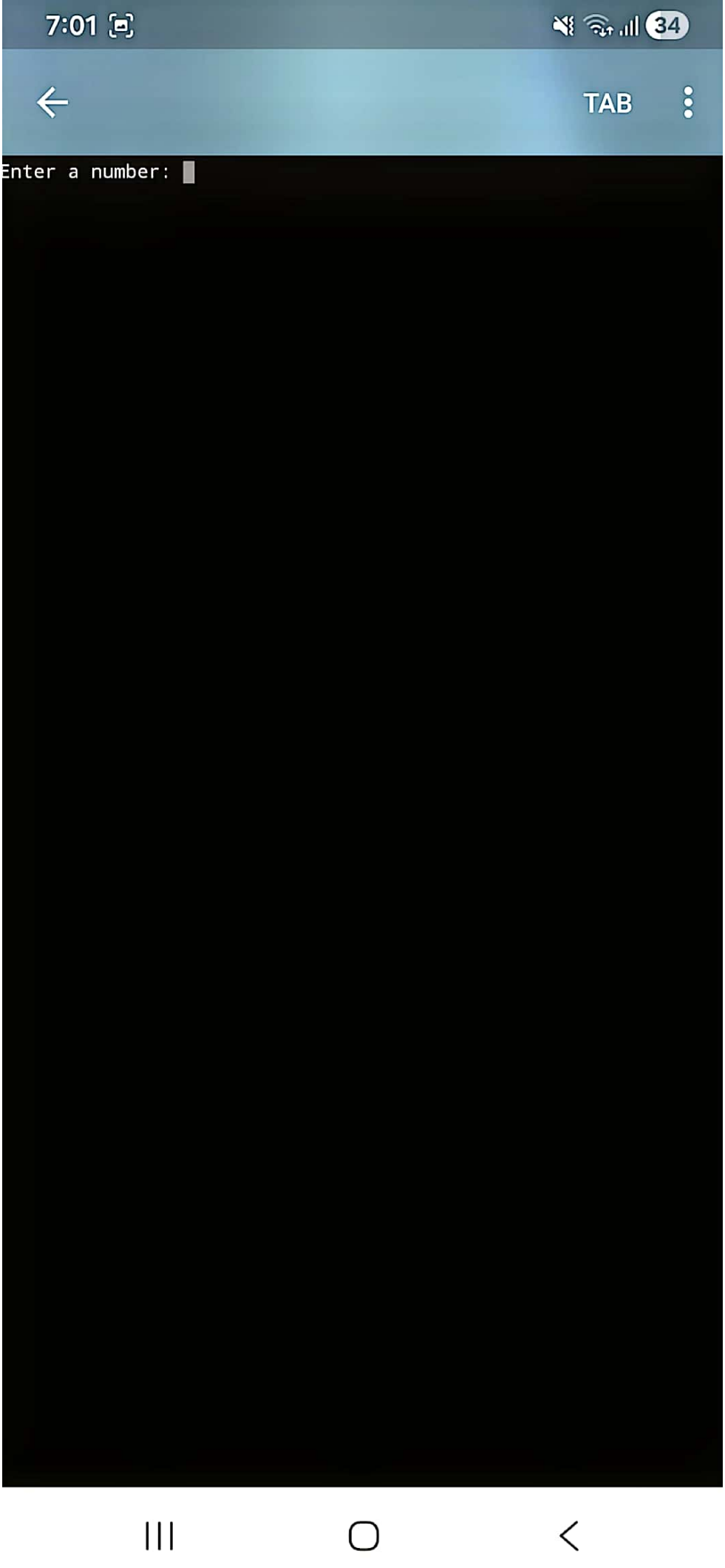
output

Enter a number

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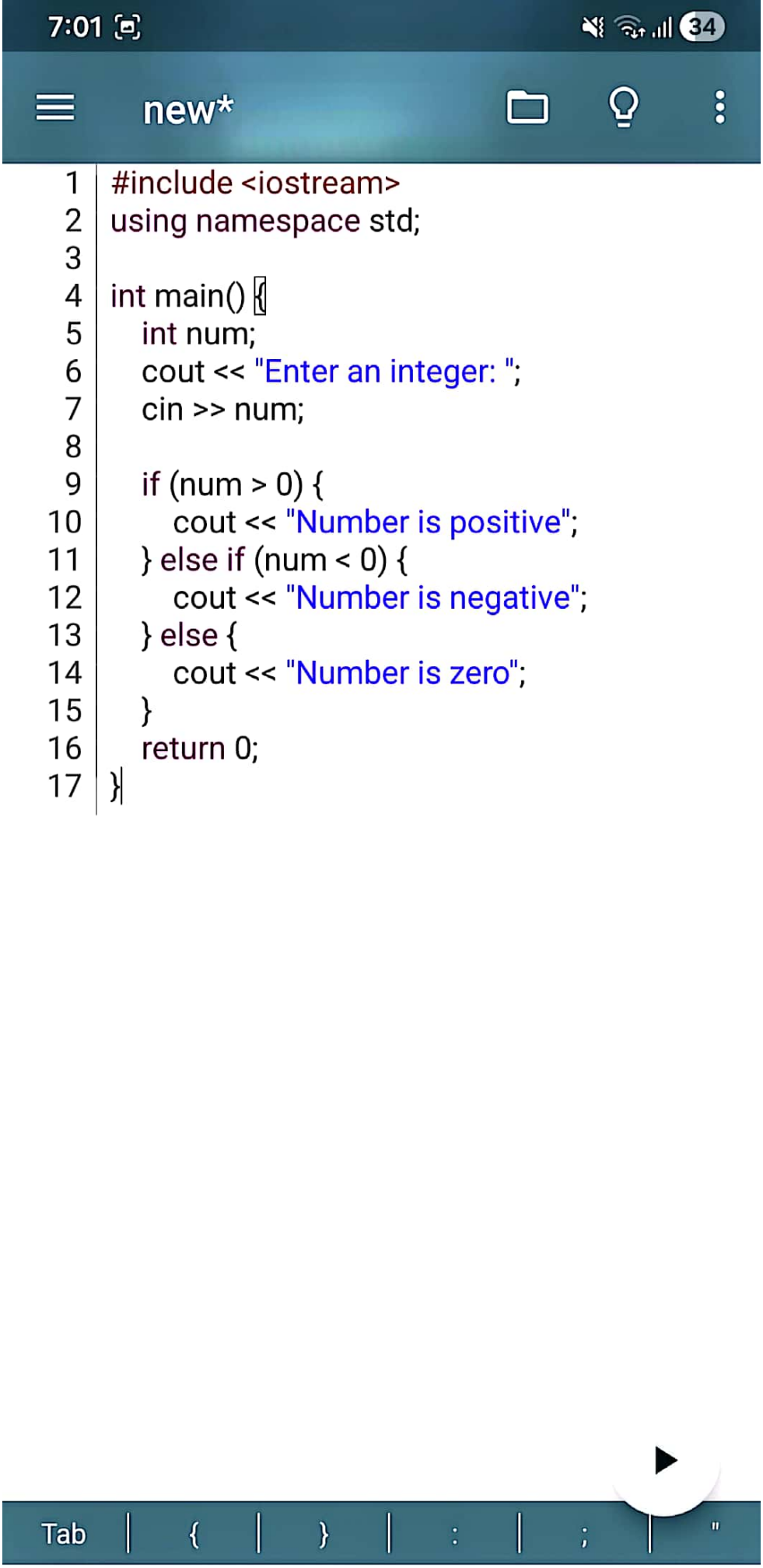
Task 2
Positive, Negative or Zero

```
#include <iostream>
using namespace std;
int main(){
    int num;
    if(num>0){
        cout<<"Number is positive";
    }else if(num<0){
        cout<<"Number is negative";
    }else{
        cout<<"Number is zero";
    }
    return 0;
}
```

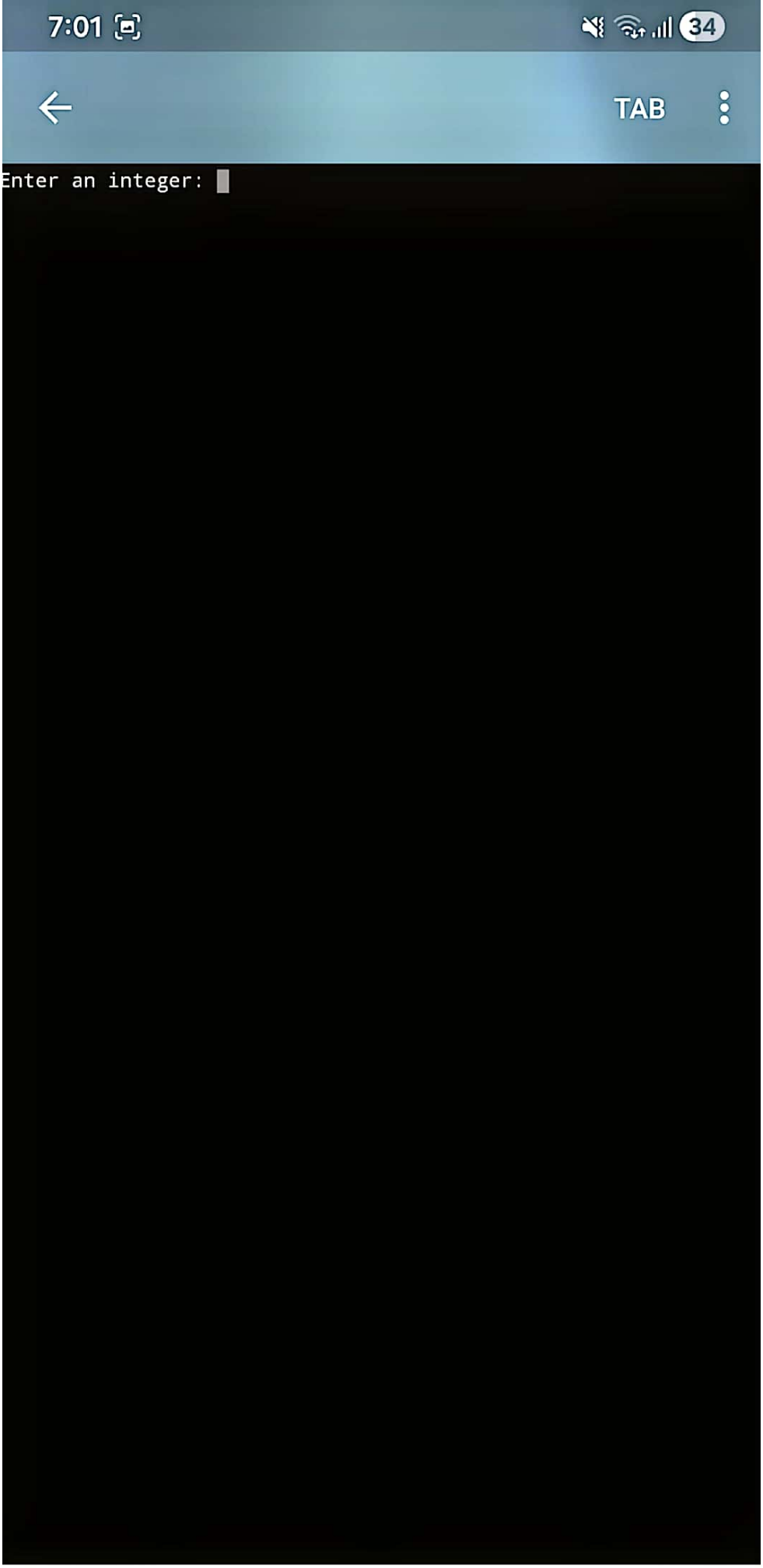
output

Enter an integer

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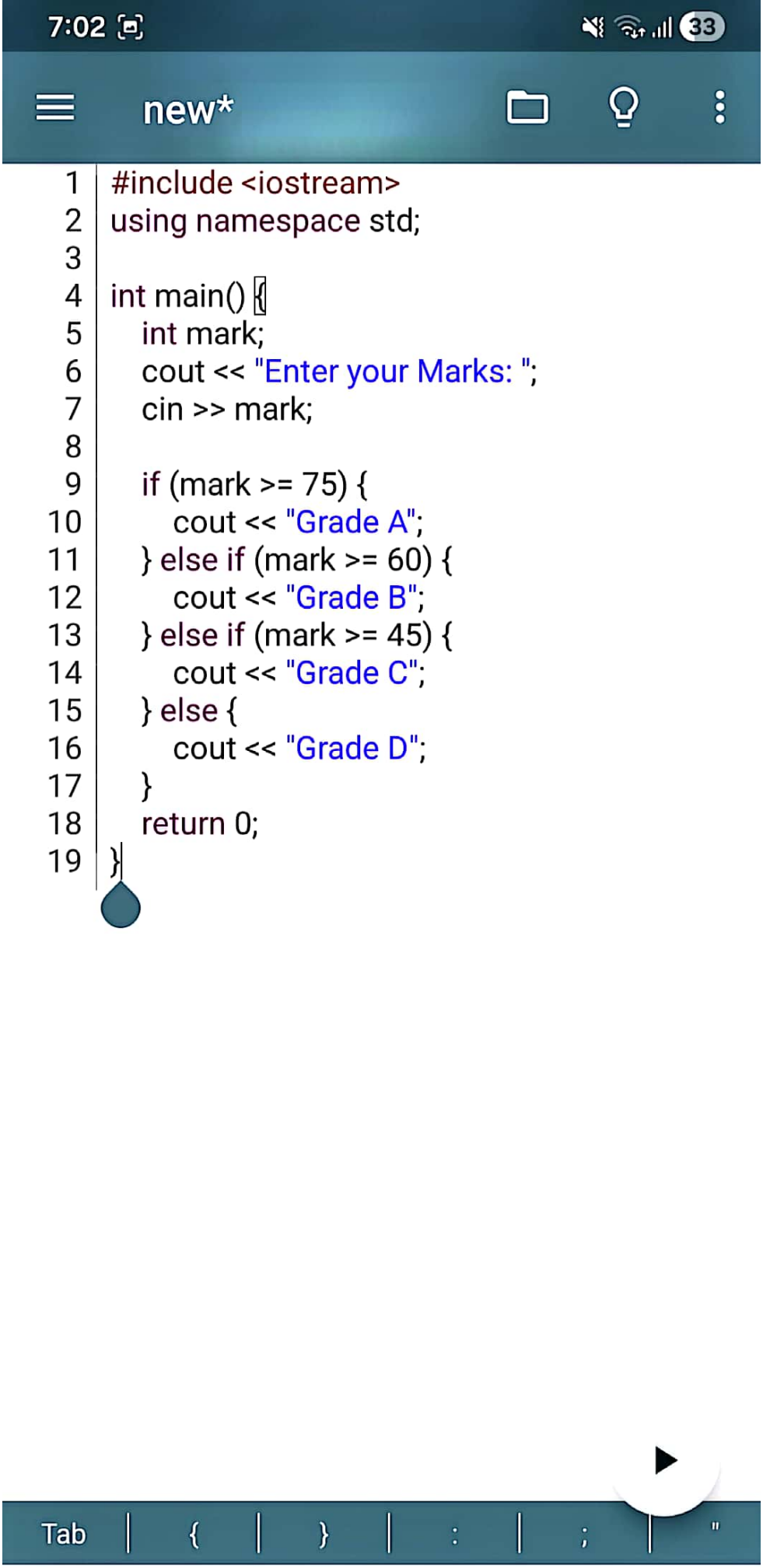
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Task 3
Grade system

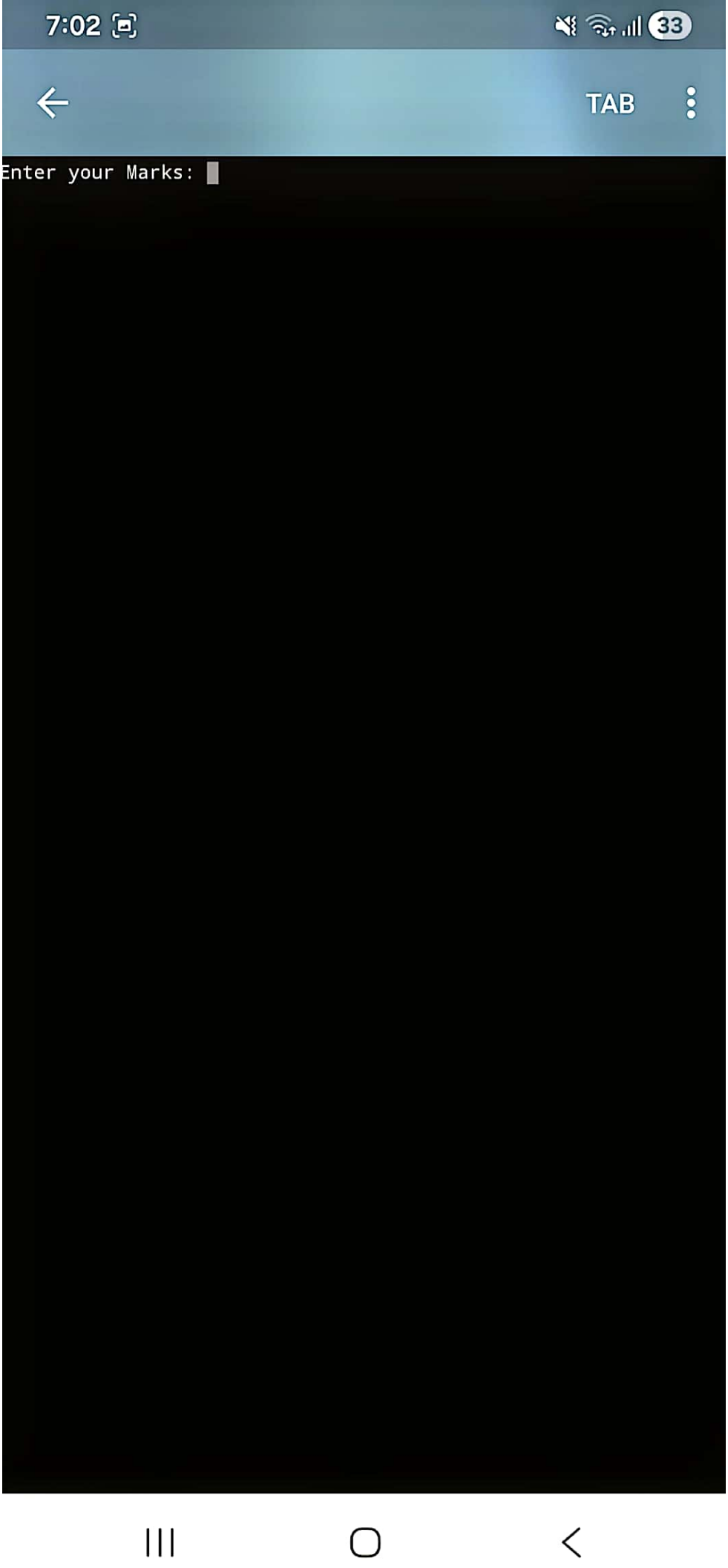
```
#include <iostream>
using namespace std;
int main() {
    int mark;
    cout << "Enter your marks: ";
    cin >> mark;
    if (mark >= 75) {
        cout << "Grade A";
    } else if (mark >= 60) {
        cout << "Grade B";
    } else if (mark >= 45) {
        cout << "Grade C";
    } else {
        cout << "Grade D";
    }
    return 0;
}
```

output= Enter your marks

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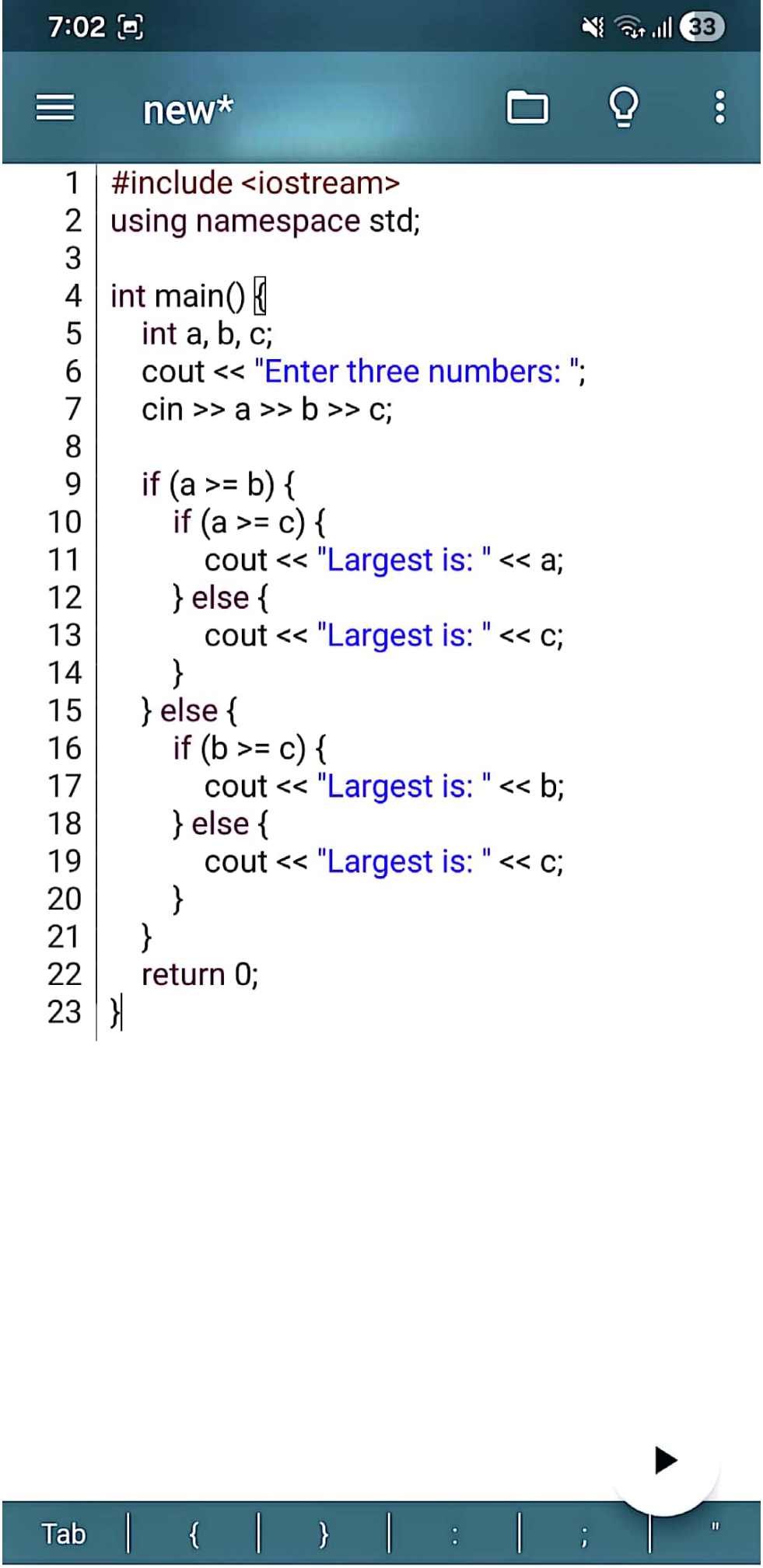
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Task 4
Largest Number + Digits count

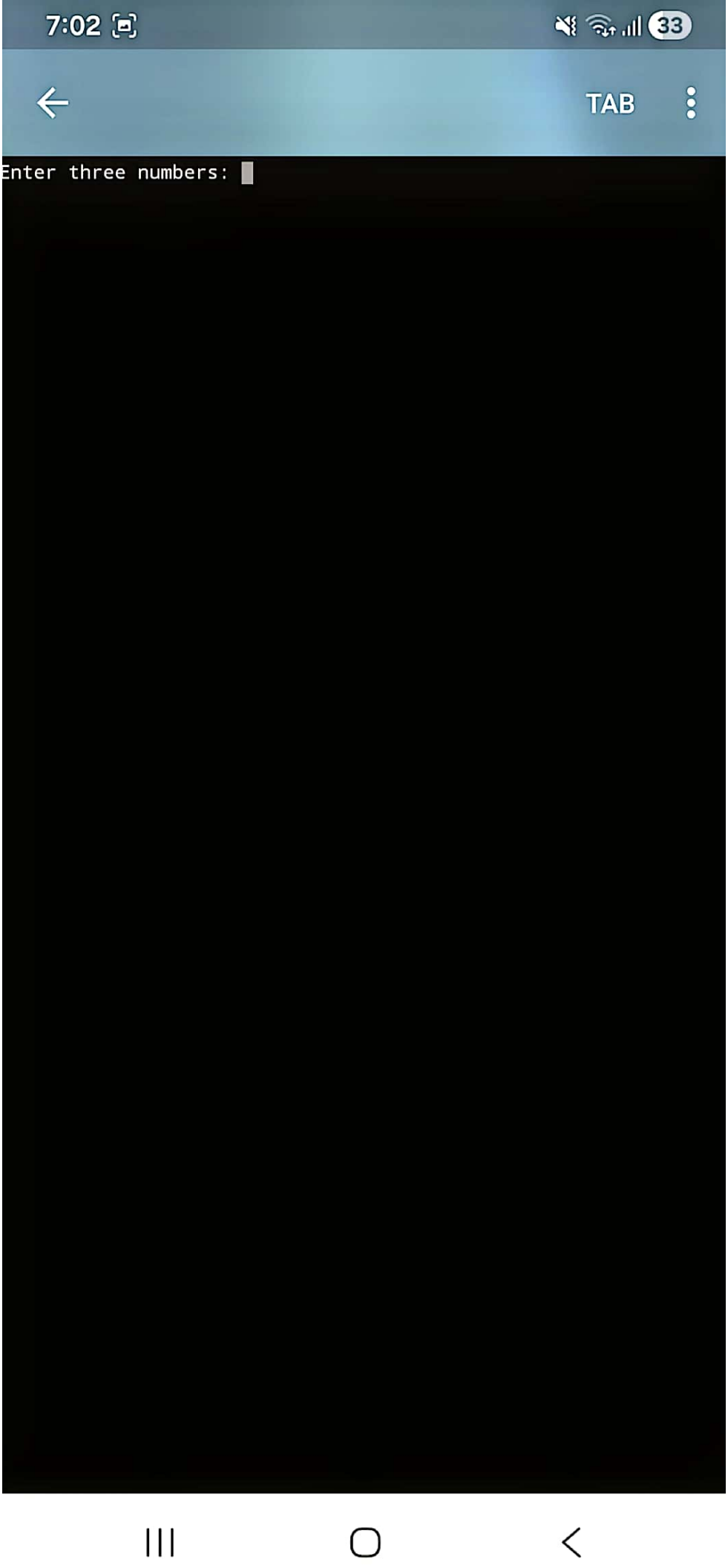
A) Largest Number using Nested if..else

```
#include<iostream>
using namespace std;
int main( ){
    int a,b,c;
    cout<<"Enter three numbers:";
    cin>>a>>b>>c;
    if (a>=b){
      if(a>=c){
        cout<<"Largest is="<<a;
    } else {
        cout<<"Largest is="<<c;
    }
  } else {
        if (b>=c){
          cout<<"Largest is: "<<b;
        }else{
          cout<<"Largest is="<<c;
        }
    } return 0;
}
```

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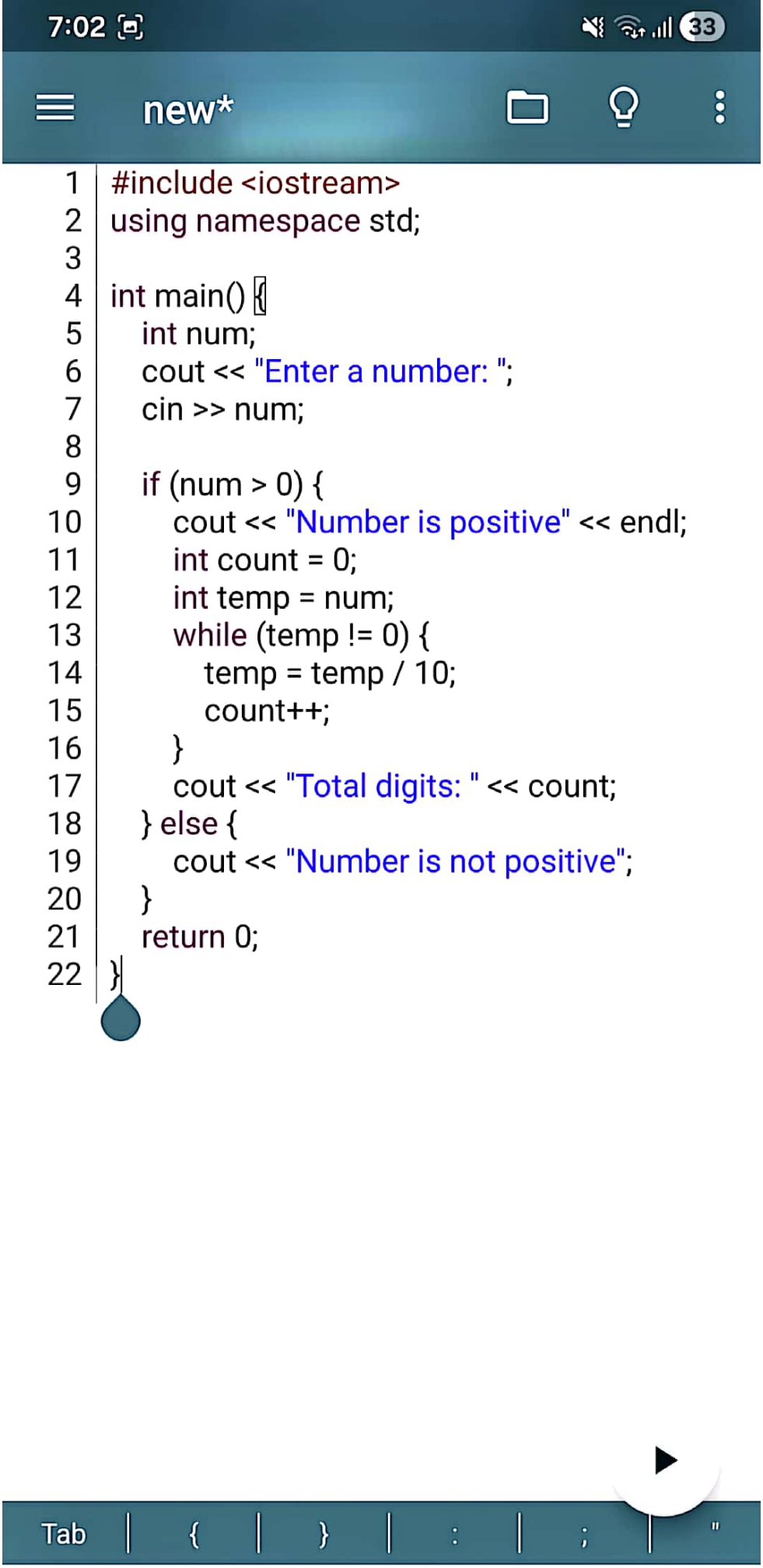
B) Positive check + Digits count

```
#include<iostream>
using namespace std;
int main(){
    int num;
    cout<<"Enter a number:";
    cin>>num;
    if(num>0){
        cout<<"Number is positive"<<endl;
        int count=0;
        int temp=num;
        while(temp!=0){
            temp=temp/10;
            count++;
        }
        cout<<"Total digits:"<<count;
    }else{
        cout<<"Number is not positive";
    }
    return 0;
}
```

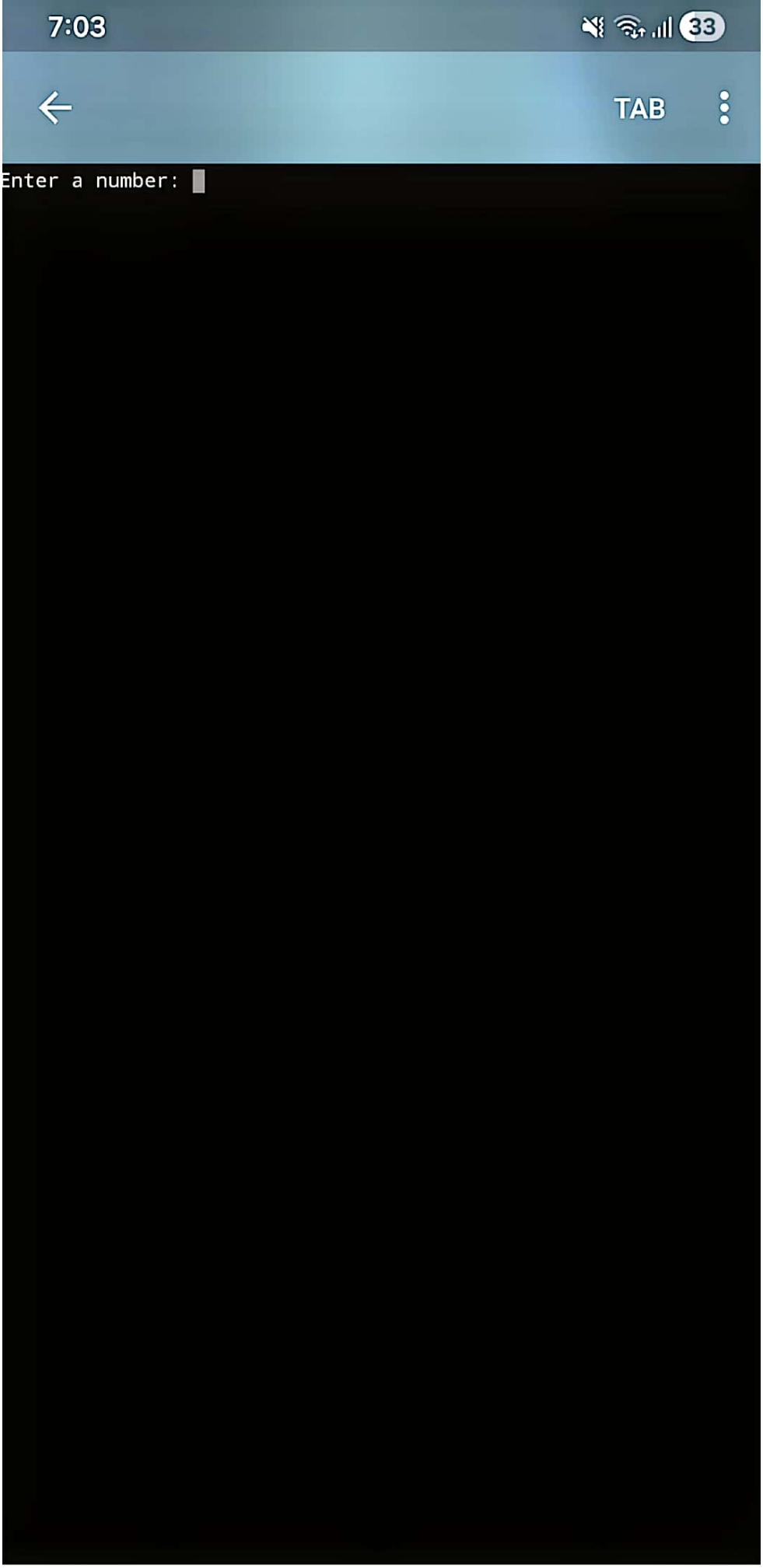
output:

Enter a number

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